

n8n Background Workflows

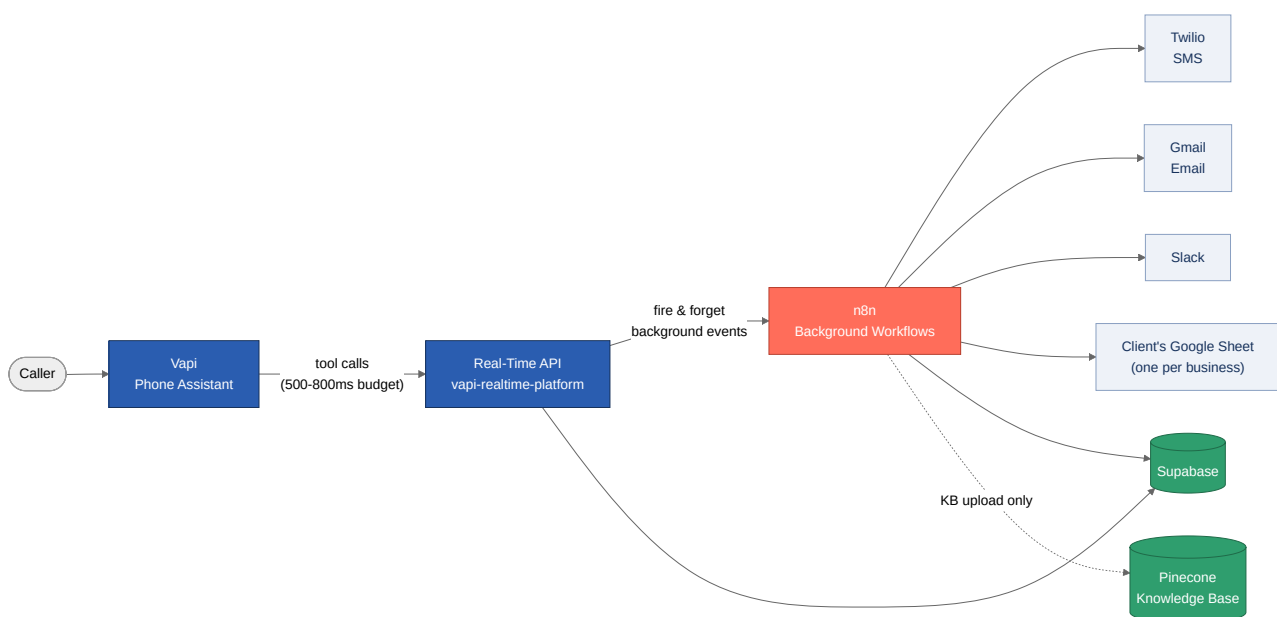
Setup Guide — plain language, with diagrams

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n8n Background Workflows — Setup Guide (plain-language)

Looking for the full start-to-finish process (database → real-time API → Vapi assistant → phone number → **these workflows** → testing)? See [../docs/MASTER-Implementation-Guide.pdf](#), which includes everything below as one of its phases. This file is the focused, n8n-only reference.

This folder has **9 workflows**. They run *after* the phone call already happened — nothing here talks to the caller in real time. Import them into n8n in the order listed below.



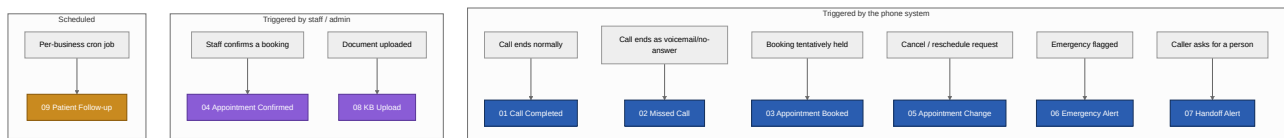
The phone system (Vapi) only ever talks to the real-time API. The API answers the caller immediately, then separately — without making the caller wait — tells n8n what happened. n8n is the only thing that talks to Twilio, Gmail, Slack, each client's Google Sheet, and (for knowledge uploads) Pinecone.

What changed from your original 5 files

- **Appointment Booked** no longer texts the customer "confirmed." Every phone booking is a *tentative hold* — the customer gets "we've received your request, our team will confirm shortly," and your staff get a Slack ping to review it.
- **New: Appointment Confirmed.** This is the workflow that sends the real "you're confirmed" text/email — but only when a staff member actually triggers it (see "How staff confirm a booking" below). This matches what you asked for: staff confirm, then the confirmation goes out.
- **New: Emergency Alert, Handoff Alert, Appointment Change (cancel/reschedule).** These three didn't exist before. They cover the AI phone system flagging an emergency, a caller asking for a real person, and a cancel/reschedule request — all of which previously had nowhere to notify your team.

- **GoHighLevel (GHL) has been replaced with Google Sheets — one spreadsheet per client.** Every note/task that used to go to GHL now writes a row to that client's own Google Sheet instead (see the tab schema below). This only changes background, after-the-call logging — nothing about the live phone call changes.
- **Call history now lives in each client's Sheet, not Supabase.** The `Call_Log` tab replaces the old `call_transcripts /detailed-calls` writes. (A small internal record still exists in Supabase automatically — see "What still lives in Supabase" below — but the transcript, summary, and sentiment your staff actually read now live in the client's Sheet.)
- **Knowledge Base Upload** now has the same safety net as your other workflows: if something fails partway through, it gets logged and posted to `#automation-alerts` instead of failing silently.
- Every workflow that writes to a client's Sheet now **checks first** whether that business has one configured. If it doesn't, it skips that one step instead of erroring out — nothing else in the workflow is affected.
- Field names in every workflow now match exactly what the phone-answering API (`vapi-realtime-platform`) sends. No manual re-mapping needed.

The 9 workflows, in import order



#	File	Fires when	What it does
1	01-call-completed.json	Every call ends (normal conversation)	Logs the call to the client's Sheet, AI-summarizes it, pings staff if it needs follow-up
2	02-missed-call.json	A call ends as voicemail/no-answer	Texts the caller back, emails/Slacks staff, logs it to the client's Sheet
3	03-appointment-booked.json	AI tentatively holds a time slot	Texts customer "request received," logs it to the client's Sheet, pings staff to confirm
4	04-appointment-confirmed.json	Staff confirm a booking	Sends the real "you're confirmed" text + email, updates the Sheet row and your records
5	05-appointment-change.json	Caller cancels/reschedules	If it's straightforward: confirms the change. If it's last-minute or unclear: tells the customer staff will follow up, and pings staff urgently
6	06-emergency-alert.json	AI flags an emergency	Texts + emails your on-call number immediately, logs it to the client's Sheet, Slacks the team
7	07-handoff-alert.json	Caller asks for a real person	Emails staff, logs it to the client's Sheet, Slacks the team
8	08-kb-upload.json	You upload/update a knowledge document	Splits it into chunks, creates embeddings, stores it in Pinecone, tells the phone API to stop using the old cached answers
9	09-patient-followup.json	Scheduled (per business)	Sends review-request texts a few days after a completed visit

Import 1–8 as regular workflows (each has its own webhook trigger). Workflow 9 is a bit different — see below.

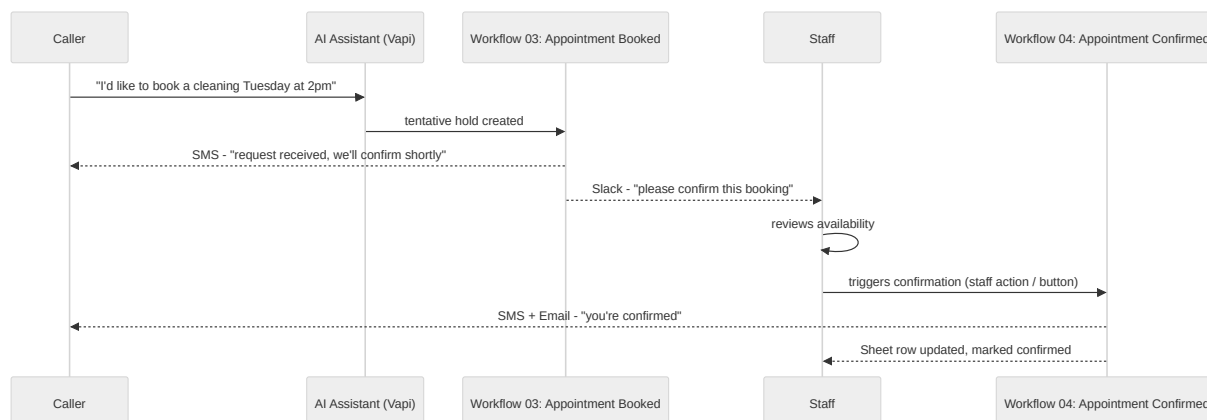
Each client's Google Sheet: the tabs you need to create

Create **one Google Sheet per client** (matches your original build's pattern), and put its ID (from the sheet's URL) into that business's `google_sheet_id` field. Add these tabs, with headers in row 1 exactly as named:

Tab name	Written by	Columns
Call_Log	Workflow 1	call_id , timestamp , customer_name , customer_phone , recording_url , duration_sec , transcript , summary , sentiment , action_items
Missed_Calls	Workflow 2	timestamp , call_id , caller_phone , reason , voicemail_url , staff_notified
Bookings	Workflows 3 & 4	appointment_id , timestamp , customer_name , customer_phone , service , start_time , status , confirmed_at
Appointment_Changes	Workflow 5	timestamp , customer_name , customer_phone , action , status , reason , new_time
Emergencies	Workflow 6	timestamp , caller_phone , issue_summary , severity , action_taken
Handoffs	Workflow 7	timestamp , caller_phone , caller_email , reason

`Bookings` is written by two different workflows on purpose: workflow 3 creates the row when the AI tentatively holds a slot (`status = tentative`), and workflow 4 updates that *same row* (matched by `appointment_id`) once staff confirm (`status = confirmed`) — so you always see one row per appointment, not two.

How staff confirm a booking (workflow 4)

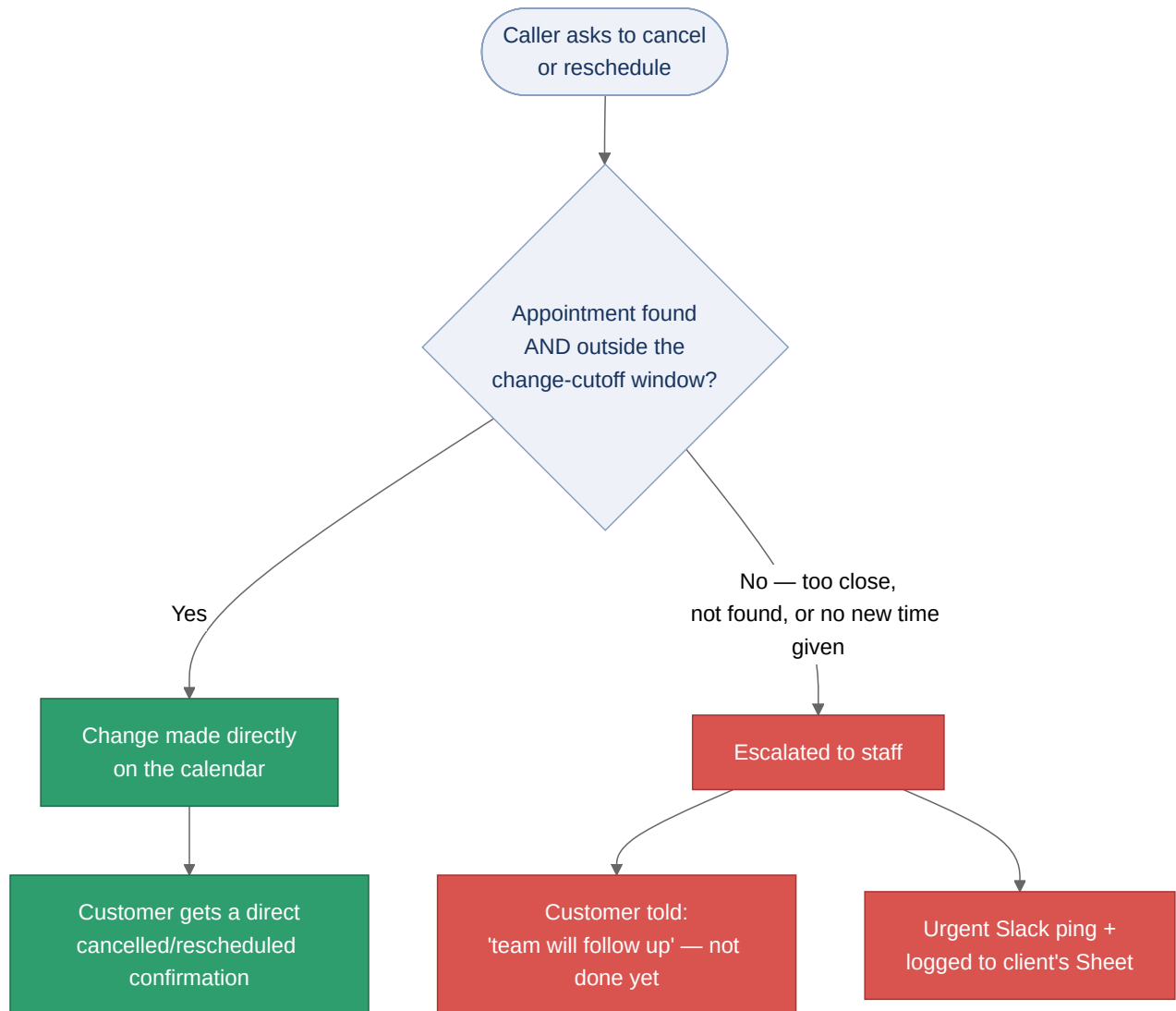


Workflow 4 needs *something* to call it once a staff member has actually checked and confirmed the appointment. Pick whichever is easiest for you — none of these require writing code:

- **Simplest:** a GoHighLevel automation (if you still use GHL for pipeline management) that fires this webhook whenever an appointment's stage is moved to "Confirmed" — you can keep GHL for staff-side pipeline tracking even though it's no longer where notes/tasks get written.
- **Also simple:** any basic web form or button (even a bookmarked link) your front desk clicks that sends the appointment's details to this webhook.

Whichever you pick, it needs to send the same information as workflow 3 (customer name/phone/email, appointment time, service, business name, sheet ID).

How cancel/reschedule decisions work (workflow 5)



The phone system only makes the change itself when it's confident and there's enough notice — otherwise it hands it to staff rather than guessing.

Credentials you'll need in n8n

Credential	Used by	Notes
Supabase	most workflows	same database as the rest of the project (internal logs, not client-visible history)
Google Sheets OAuth2	Call Completed, Missed Call, Appointment Booked, Appointment Confirmed, Appointment Change, Emergency Alert, Handoff Alert	connect a Google account with edit access to every client's spreadsheet (share each client Sheet with that account, or use one shared Drive)
Gmail	Call Completed, Missed Call, Appointment Confirmed, Emergency Alert, Handoff Alert	staff-facing emails
Twilio	Missed Call, Appointment Booked, Appointment Confirmed, Appointment Change, Emergency Alert, Patient Follow-up	customer-facing texts
Slack	every workflow	team notifications — see channel list below
OpenAI	Call Completed (summarizing), KB Upload (embeddings)	
Pinecone	KB Upload	your knowledge base storage
Internal API Key (HTTP Header Auth)	KB Upload's "Flush Knowledge Cache" step	header name <code>x-internal-secret</code> , value must match the phone API's <code>INTERNAL_API_SHARED_SECRET</code> setting

Environment variables n8n needs

Set these in n8n's environment (Settings → Environment Variables, or your hosting platform's config):

- `TWILIO_FROM_NUMBER` — the number texts are sent from
- `FRONT_DESK_ALERT_EMAIL` — where staff email alerts go
- `FRONT_DESK_ALERT_PHONE` — where the Emergency Alert text goes (an on-call phone number)
- `REALTIME_API_BASE_URL` — the phone-answering API's base URL (used only by KB Upload's cache-flush step)

There's no environment variable for which Sheet belongs to which client — that comes through automatically in each event's `sheetId` field, sourced from that business's `google_sheet_id` setting in the phone API's database.

Default Slack channels (override anytime)

Every workflow accepts a `slackChannel` field from the caller; if it's blank, these defaults are used:

`#calls`, `#missed-calls`, `#bookings` (both booked and confirmed), `#appointment-changes`, `#emergencies`, `#handoffs`, `#kb-updates`, `#followups`. All error alerts always go to `#automation-alerts` regardless.

What still lives in Supabase

Client-visible call/booking/emergency/handoff history now lives in each client's Google Sheet (above) — that's what your staff should actually read. Supabase still holds a few things these workflows use internally, which your staff never need to open directly: `follow_up_tasks` (internal to-do list, separate from the client-facing Sheet), `appointments` (the phone system's own booking records), `kb_document_versions`, `audit_log`, `workflow_execution_log`, `workflow_errors`. The phone-answering API also keeps a minimal internal `calls` record (call id, timing, business) for its own correlation/analytics — this is not the same as the rich `Call_Log` tab in the client's Sheet, and isn't meant to be read directly.

A quick word on the "Has Sheet ID?" checks

If a business hasn't had a Google Sheet set up yet (`google_sheet_id` blank), these workflows now simply **skip** the Sheets-logging step and carry on with everything else (texts, emails, Slack). Nothing fails or gets stuck — it just means that one business won't have a browsable history until you create and connect their Sheet.

Testing checklist before going live

1. Book a test appointment through the phone system → confirm the customer gets a "we've received your request" text (not "confirmed"), staff get a Slack ping, and a `tentative` row appears in `Bookings`.
2. Trigger workflow 4 manually with test data → confirm the customer now gets the real "confirmed" text + email, and the same `Bookings` row updates to `confirmed` (not a second row).
3. Ask the AI to cancel a test appointment close to its start time → confirm staff get an urgent Slack ping, the customer is told staff will follow up (not "cancelled"), and a row appears in `Appointment_Changes` with `status = escalated`.
4. Cancel a test appointment far in advance → confirm the customer gets a direct "cancelled" confirmation and the `Appointment_Changes` row shows `status = done`.
5. Simulate an emergency call → confirm the on-call phone/email/Slack all fire and a row appears in `Emergencies`.
6. Ask for "a real person" → confirm the handoff email/Slack fire and a row appears in `Handoffs`.

7. Upload a test knowledge document → confirm it appears in Pinecone and the `#kb-updates` Slack message posts; then try breaking it (bad file) to confirm `#automation-alerts` fires instead of the workflow silently failing.
8. Let a test call go to voicemail → confirm the callback text fires and a row appears in `Missed_Calls`.
9. Remove a test business's `google_sheet_id` temporarily → confirm every workflow still completes (texts/emails/Slack still fire), just without a Sheets row — nothing should error.